ECON 122

MEMORANDUM
Exam scope 2026 · Opportunity 1 style

Module code	ECON122
Module description	Introduction to Macroeconomics
Type of assessment	Memorandum: Comprehensive revision examination (Paper 1 style)
Maximum marks	500
Structure	Section A: Multiple choice [200] | Section B: Theory [150] | Section C: Calculations [150]
Suggested time	15 hours in total (≈ 3 hours per 100 marks, as in the real exam). Work in sittings: Section A (6 h), Section B (4½ h), Section C (4½ h).
Textbook	Mohr & Associates, *Economics for South African Students* – Chapters 13–18 and 20–22
Resources	Non-programmable calculator

This is an independent practice paper written to mirror the format, mark allocation and style of past ECON122 Paper 1 examinations (2018–2024) and the 2026 exam scope. It is not an official university examination paper. All companies, countries and data are fictional or hypothetical unless stated otherwise.

SECTION A — QUICK ANSWER KEY — 2 marks each

1. A	**2.** D	**3.** B	**4.** D	**5.** A
6. B	**7.** A	**8.** D	**9.** B	**10.** A
11. D	**12.** B	**13.** B	**14.** C	**15.** D
16. A	**17.** B	**18.** D	**19.** B	**20.** B
21. C	**22.** D	**23.** B	**24.** B	**25.** C
26. A	**27.** B	**28.** B	**29.** C	**30.** C
31. B	**32.** C	**33.** C	**34.** D	**35.** B
36. B	**37.** D	**38.** C	**39.** D	**40.** D
41. A	**42.** D	**43.** A	**44.** D	**45.** B
46. C	**47.** C	**48.** D	**49.** A	**50.** B
51. C	**52.** A	**53.** B	**54.** B	**55.** B
56. B	**57.** B	**58.** C	**59.** B	**60.** B
61. A	**62.** C	**63.** D	**64.** D	**65.** D
66. A	**67.** C	**68.** B	**69.** A	**70.** D
71. A	**72.** D	**73.** D	**74.** A	**75.** C
76. D	**77.** C	**78.** D	**79.** A	**80.** A
81. A	**82.** D	**83.** D	**84.** D	**85.** A
86. A	**87.** C	**88.** A	**89.** A	**90.** A
91. C	**92.** A	**93.** A	**94.** C	**95.** C
96. C	**97.** C	**98.** C	**99.** C	**100.** C

How to mark yourself

- Section A: 2 marks per correct answer; each answer below includes a short explanation.
- Sections B and C: marks are shown in brackets after each step, e.g. (½) or (1). In calculations, NWU markers usually award ½ for the formula, marks for correct substitution and ½–1 for the final answer rounded to two decimals – so always write the formula.
- Theory answers list the key points a marker looks for. Equivalent wording in your own words earns the marks.
- Score guide: 75%+ (375+) exam ready · 60–74% (300–374) revise weak chapters · below 60% go back to the study guide exercises.

SECTION A / *AFDELING A* — MULTIPLE CHOICE [200]

Choose the **most correct** answer. Each question carries 2 marks (100 × 2 = 200).

1. Gross domestic product (GDP) is the total value of all ___ produced within the boundaries of a country in a particular period.

A. final goods and services
B. final goods sold to households only
C. goods and services produced by permanent residents, wherever they live
D. goods and services, including intermediate goods

Answer: A — GDP counts only final goods and services to avoid double counting.

2. Which one of the following would be included in South Africa's GDP for 2026?

A. The purchase of existing shares on the JSE
B. The sale of a 2019 second-hand car in 2026
C. The salary of a South African engineer working in Dubai
D. The value of a new house built in Polokwane during 2026

Answer: D — Only current production within SA's borders counts; resales and financial transactions are excluded.

3. GDP at market prices is equal to GDP at factor cost plus ...

A. depreciation (consumption of fixed capital)
B. taxes on production and imports minus subsidies
C. net primary income from the rest of the world
D. subsidies minus taxes on production and imports

Answer: B — GDPmp = GDPfc + taxes - subsidies.

4. GDP at factor cost is R4 800 billion, taxes on production and imports are R620 billion and subsidies are R45 billion. GDP at market prices is:

A. R4 135 billion
B. R4 225 billion
C. R5 465 billion
D. R5 375 billion

Answer: D — 4 800 + 620 - 45 = R5 375 billion.

5. Given C = R3 200 bn, I = R900 bn, G = R1 100 bn, X = R1 500 bn and Z = R1 450 bn, gross domestic expenditure (GDE) is:

A. R5 200 bn
B. R3 750 bn
C. R6 700 bn
D. R5 250 bn

Answer: A — GDE = C + I + G = 3 200 + 900 + 1 100 = R5 200 bn (exports and imports are excluded).

6. Which statement best explains why South Africa's gross national income (GNI) is usually smaller than its GDP?

A. South Africa imports more goods than it exports
B. South Africa pays more primary income to the rest of the world than it receives
C. Taxes on production exceed subsidies
D. Depreciation of the capital stock is very large

Answer: B — GNI = GDP + primary income receipts - primary income payments; SA's payments exceed its receipts.

7. A farmer sells wheat for R10 000 to a miller, who sells flour for R12 500 to a baker, who sells bread for R18 000 to a shopkeeper, who sells it to consumers for R21 000. The value added by the baker is:

A. R5 500 **B.** R2 500
C. R18 000 **D.** R3 000

Answer: A — Value added = sales - purchases = 18 000 - 12 500 = R5 500.

8. Real GDP measures the value of production ...

A. at factor cost only **B.** after subtracting depreciation
C. at current prices **D.** at constant (base-year) prices

Answer: D — Real GDP removes the effect of price changes by using base-year prices.

9. Real GDP was R4 400 billion in 2025 and R4 532 billion in 2026. The real economic growth rate for 2026 is:

A. 2.91% **B.** 3.00%
C. 13.20% **D.** 30.00%

Answer: B — (4 532 - 4 400) / 4 400 x 100 = 3.00%.

10. When GDP is calculated using the income approach, it is valued at:

A. factor cost **B.** basic prices
C. market prices **D.** constant prices

Answer: A — Income method = factor cost; production method = basic prices; expenditure method = market prices.

11. When GDP is calculated using the production (value-added) approach, it is valued at:

A. current exchange rates **B.** market prices
C. factor cost **D.** basic prices

Answer: D — The production method uses basic prices.

12. The further the Lorenz curve lies from the line of perfect equality (the diagonal), ...

A. the lower the Gini coefficient
B. the greater the inequality in the income distribution
C. the more equal the income distribution
D. the higher the per capita income

Answer: B — Distance from the diagonal measures inequality.

13. A Gini coefficient of 0 indicates ...

A. that the Lorenz curve lies on the horizontal axis
B. perfect equality
C. perfect inequality
D. that one household earns all income

Answer: B — Gini ranges from 0 (perfect equality) to 1 (perfect inequality).

14. The quantile ratio as a measure of inequality compares ...

A. the area between the diagonal and the Lorenz curve with the total area
B. nominal GDP with real GDP
C. the income share of the richest group with that of the poorest group
D. the number of employed with the labour force

Answer: C — E.g. income share of top 10% / bottom 10%.

15. The current account of the balance of payments records ...

A. only changes in gold and foreign reserves
B. flows of shares, bonds and direct investment
C. only unrecorded transactions (errors and omissions)
D. exports and imports of goods and services plus income payments to and from other countries

Answer: D — Financial flows are in the financial account.

16. Which one of the following is NOT one of the five main macroeconomic objectives?

A. Maximising government tax revenue
B. Full employment
C. Price stability
D. Equitable distribution of income

Answer: A — The five objectives: growth, full employment, price stability, BoP stability, equitable income distribution.

17. A shop labels a loaf of bread at R19.99. Here money is performing its function as a ...

A. medium of exchange
B. unit of account
C. store of value
D. standard of deferred payment only

Answer: B — Prices are expressed in money: unit of account.

18. Thabo keeps R5 000 in a savings account to spend next year. Money is serving as a ...

A. medium of exchange
B. factor of production
C. unit of account
D. store of value

Answer: D — Holding purchasing power over time = store of value.

19. High inflation most directly undermines money's function as a ...

A. means of barter
B. store of value
C. unit of account
D. medium of exchange

Answer: B — Inflation erodes the purchasing power of stored money.

20. M1 in South Africa consists of ...

A. M2 plus long-term deposits
B. coins and banknotes in circulation plus demand deposits of the domestic private sector
C. coins and banknotes only
D. all short- and medium-term deposits

Answer: B — M1 = C + D (narrowest measure, medium of exchange).

21. M3, the most comprehensive measure of money, is equal to ...

A. coins and notes plus cheque deposits
B. M2 minus quasi-money
C. M2 plus long-term deposits (longer than six months)
D. M1 plus short- and medium-term deposits

Answer: C — M3 = M2 + long-term deposits.

22. Coins and notes in circulation = R50 m, demand deposits = R300 m, short- and medium-term deposits = R400 m and long-term deposits = R250 m. M2 is:

A. R1 000 m
B. R650 m
C. R350 m
D. R750 m

Answer: D — M1 = 350; M2 = 350 + 400 = R750 m.

23. The demand for active balances (transactions and precautionary motives) depends mainly on ...

A. expected bond prices only
B. the level of income
C. the interest rate
D. the exchange rate

Answer: B — Active balances (L1) are a function of income.

24. The speculative demand for money (passive balances) is ...

A. positively related to the interest rate
B. inversely related to the interest rate
C. independent of the interest rate
D. determined only by the level of income

Answer: B — Higher interest = higher opportunity cost of holding money.

25. An increase in national income will, ceteris paribus, ...

A. have no effect on the demand for money
B. cause a movement up along LL
C. shift the total demand for money curve (LL) to the right
D. shift LL to the left

Answer: C — Higher income raises demand for active balances, shifting LL right.

26. If the interest rate increases from 7% to 8%, ceteris paribus, there will be ...

A. a movement along LL to a smaller quantity of money demanded
B. a leftward shift of LL
C. a rightward shift of LL
D. an increase in the quantity of money demanded

Answer: A — A change in i causes a movement along the curve, not a shift.

27. In the South African approach to the money supply, the quantity of money is determined by ...

A. the SARB printing a fixed amount of banknotes
B. the demand for money and the interest rate (there is no independent money supply curve)
C. the amount of gold held by the SARB
D. the government budget deficit only

Answer: B — Money is demand-determined (endogenous); SARB influences it via the repo rate.

28. The main instrument of monetary policy used by the SARB is the ...

A. personal income tax rate
B. repo rate
C. VAT rate
D. import tariff

Answer: B — Accommodation policy through the repo rate.

29. When the SARB sells government bonds in the open market, it ...

A. lowers the repo rate automatically
B. increases banks' cash reserves
C. reduces banks' cash reserves and enlarges the money-market shortage
D. increases the money supply

Answer: C — Selling bonds drains liquidity.

30. Which one of the following is NOT a function of the South African Reserve Bank?

A. Acting as banker and adviser to the government
B. Acting as lender of last resort to banks
C. Collecting personal income tax from individuals
D. Formulating and implementing monetary policy

Answer: C — Tax collection is SARS's function.

31. According to your study material, the inflation target range adopted in South Africa in 2000 is:

A. 0-3%
B. 3-6%
C. 6-9%
D. 1-5%

Answer: B — 1-5% was the original 1998 proposal; 3-6% from 2000.

32. Public goods are goods that are ...

A. non-rival but excludable
B. rival but non-excludable
C. non-rival and non-excludable
D. rival and excludable

Answer: C — E.g. national defence, street lighting.

33. The free-rider problem associated with public goods arises mainly because public goods are ...

A. rival in consumption
B. subject to VAT
C. non-excludable
D. always produced by monopolies

Answer: C — Non-payers cannot be excluded, so people avoid paying.

34. Fish in the open ocean are an example of a ...

A. natural monopoly
B. pure public good
C. merit good
D. common property resource

Answer: D — Rival but non-excludable: tragedy of the commons.

35. When a factory pollutes a river, ...

A. the good is underproduced
B. marginal social cost exceeds marginal private cost and the good is overproduced
C. marginal private cost exceeds marginal social cost
D. social benefit exceeds private benefit

Answer: B — Negative externality: MSC > MPC, market output too high.

36. The Coase theorem states that ...

A. government must always tax polluters
B. if the parties to an externality can bargain, they will reach an efficient outcome without government intervention
C. monopolies should be nationalised
D. public goods must be provided by the state

Answer: B — Assumptions rarely hold in practice.

37. To achieve the socially efficient output, a tax on a polluting industry should equal ...

A. the marginal private cost
B. the VAT rate
C. the firm's total profit
D. the difference between marginal social cost and marginal private cost

Answer: D — This internalises the externality.

38. A group of firms lobbies politicians for protection so that they earn higher profits at the expense of society. This is an example of ...

A. horizontal equity
B. the benefit principle
C. rent seeking
D. a positive externality

Answer: C — Rent seeking is a cause of government failure.

39. Using the budget to reduce fluctuations in output, employment and prices refers to the government's ...

A. regulation function
B. allocation function
C. distribution function
D. stabilisation function

Answer: D — Allocation = goods markets under-supply; distribution = fairness; stabilisation = macro stability.

40. Expansionary fiscal policy involves ...

A. raising the repo rate
B. reducing government spending and/or raising taxes
C. selling government bonds in the open market
D. increasing government spending and/or reducing taxes

Answer: D — Used in a recession; often increases the budget deficit.

41. The principle that people in the same economic position should pay the same amount of tax is known as ...

A. horizontal equity
B. vertical equity
C. neutrality
D. the benefit principle

Answer: A — Vertical equity: different positions, different taxes.

42. Value-added tax (VAT) in South Africa is regarded as ...

A. a direct tax
B. progressive
C. proportional
D. regressive

Answer: D — VAT takes a bigger share of income from low-income earners.

43. The legal minimisation of one's tax liability is called ...

A. tax avoidance
B. tax evasion
C. tax incidence
D. tax rebate

Answer: A — Evasion is illegal non-payment.

44. Use the 2024/25 SARS table in Section C (Question 7). A person younger than 65 has a taxable income of R420 000. The tax payable is:

A. R130 200
B. R60 237
C. R92 707
D. R75 472

Answer: D — 77 362 + 31% x (420 000 - 370 500) - 17 235 = R75 472.

45. For the person in the previous question, the effective (average) tax rate is:

A. 31.00%
B. 17.97%
C. 22.07%
D. 26.00%

Answer: B — 75 472 / 420 000 x 100 = 17.97% (22.07% forgets the rebate).

46. A country has a comparative advantage in a good when it can produce that good ...

A. at a higher price than another country
B. in greater quantities than any other good
C. at a lower opportunity cost than another country
D. with fewer resources than another country

Answer: C — Absolute = more output from same resources; comparative = lower opportunity cost.

47. Country A can produce 10 shirts or 20 pairs of shoes; Country B can produce 6 shirts or 18 pairs of shoes. Which statement is correct?

A. Country B has a comparative advantage in shirts
B. Country A has a comparative advantage in both goods
C. Country A has a comparative advantage in shirts
D. There is no basis for trade

Answer: C — Opportunity cost of 1 shirt: A = 2 shoes, B = 3 shoes.

48. Using the data in the previous question, mutually beneficial terms of trade for 1 shirt lie between ...

A. 6 and 10 pairs of shoes
B. 1 and 2 pairs of shoes
C. 0.33 and 0.5 pairs of shoes
D. 2 and 3 pairs of shoes

Answer: D — Between the two domestic opportunity costs.

49. If the export price index is 115 and the import price index is 125, the terms of trade index is:

A. 92.00 (the terms of trade are unfavourable)
B. 240.00
C. 108.70 (the terms of trade are favourable)
D. 92.00 (the terms of trade are favourable)

Answer: A — TOT = 115 / 125 x 100 = 92.00 < 100.

50. Which of the following represents a DEMAND for US dollars in the South African foreign exchange market?

A. An American investor buying shares on the JSE
B. A South African firm paying for machinery imported from the USA
C. An American tourist exchanging dollars for rand in Cape Town
D. A South African exporter receiving dollars for wine sold in the USA

Answer: B — SA importers need dollars.

51. Which of the following represents a SUPPLY of US dollars in the South African foreign exchange market?

A. A South African investor buying shares in an American company
B. A South African student paying tuition at a US university
C. An American tourist exchanging dollars for rand in Cape Town
D. A South African importer buying US software

Answer: C — Holders of dollars who want rand supply dollars.

52. The exchange rate changes from $1 = R17 to $1 = R19. This means that ...

A. the rand has depreciated against the dollar
B. the rand has appreciated against the dollar
C. the dollar has depreciated against the rand
D. South African exports have become more expensive in the USA

Answer: A — More rand is needed to buy one dollar.

53. A depreciation of the rand will, ceteris paribus, ...

A. make SA exports more expensive abroad
B. make SA exports cheaper abroad and imports more expensive in SA
C. make imports cheaper in SA
D. reduce the cost of imported oil

Answer: B — May improve the current account but cause imported inflation.

54. Which one of the following could cause an appreciation of the rand against the dollar?

A. An expected depreciation of the rand
B. A sharp rise in the world prices of gold and platinum
C. An increase in SA imports from the USA
D. SA investors buying US shares

Answer: B — Higher export earnings raise the supply of dollars.

55. To counter a depreciation of the rand, the SARB could ...

A. lower the repo rate
B. sell dollars from its foreign reserves
C. buy dollars in the foreign exchange market
D. remove all exchange controls

Answer: B — Selling dollars increases the supply of dollars.

56. An ad valorem tariff is ...

A. a subsidy to exporters
B. a tax levied as a percentage of the value of the imported good
C. a fixed amount per unit imported
D. a physical limit on the quantity imported

Answer: B — Specific tariff = fixed amount per unit.

57. A physical restriction on the quantity of a good that may be imported is called a(n) ...

A. export subsidy
B. import quota
C. exchange rate policy
D. ad valorem tariff

Answer: B — Quotas directly control the volume of imports.

58. The argument that a new domestic industry should be protected temporarily until it achieves economies of scale is the ...

A. dumping argument
B. government revenue argument
C. infant industry argument
D. national security argument

Answer: C — Temporary protection until competitive.

59. Selling goods in a foreign market at artificially low prices (below cost or below the home price) is called ...

A. import substitution
B. dumping
C. quota rent
D. free trade

Answer: B — Anti-dumping duties protect local firms.

60. An item costs EUR400 in Germany. If EUR1 = R20.25, the item costs in rand (ignoring transport costs):

A. R5 062.50
B. R8 100.00
C. R420.25
D. R19.75

Answer: B — 400 x 20.25 = R8 100.00.

61. When income increases by R1 000 consumption increases by R750. The marginal propensity to consume is:

A. 0.75
B. 750
C. 0.25
D. 1.33

Answer: A — c = dC / dY = 750 / 1 000.

62. Autonomous consumption refers to consumption ...

A. that changes as income changes
B. by the government sector
C. that takes place even when income is zero, financed from past savings or credit
D. of imported goods only

Answer: C — It is the intercept of the consumption function.

63. In a simple closed economy without government, if the marginal propensity to save is 0.25 the multiplier is:

A. 0.25 **B.** 1.33
C. 2.5 **D.** 4

Answer: D — alpha = 1 / (1 - c) = 1 / s = 1 / 0.25 = 4.

64. If C = 200 + 0.8Y, the savings function is:

A. S = 200 + 0.2Y **B.** S = 200 - 0.8Y
C. S = -200 + 0.8Y **D.** S = -200 + 0.2Y

Answer: D — S = Y - C; autonomous saving is negative (dissaving).

65. In the Keynesian model, equilibrium income is reached where ...

A. government spending equals tax revenue
B. savings equal zero
C. consumption equals investment
D. aggregate expenditure equals total income (A = Y), on the 45-degree line

Answer: D — Equilibrium condition Y = A.

66. In the simple Keynesian model, investment spending is treated as ...

A. autonomous: determined by interest rates and expectations, not by income
B. equal to savings at every level of income
C. negatively related to income
D. induced: a fixed fraction of income

Answer: A — Hence the investment function is a horizontal line.

67. Which one of the following is NOT an assumption of the simple Keynesian model?

A. Prices, wages, the money stock and interest rates are given
B. Initially there are only households and firms
C. Prices and wages are fully flexible
D. Spending is the driving force in the economy

Answer: C — Prices are assumed given (constant).

68. If C = 50 + 0.75Y and I = 100 in a closed economy without government, equilibrium income is:

A. 150 **B.** 600
C. 200 **D.** 450

Answer: B — Y = (50 + 100) / (1 - 0.75) = 600.

69. In the economy in the previous question, investment increases by 20. The change in equilibrium income is:

A. 80 **B.** 20
C. 26.67 **D.** 15

Answer: A — dY = 4 x 20 = 80.

70. In an open economy with c = 0.8, t = 0.25 and m = 0.1, the value of the multiplier is:

A. 5 **B.** 1.67

C. 2.5 **D.** 2

Answer: D — 1 / [1 - 0.8(0.75) + 0.1] = 1 / 0.5 = 2.

71. An increase in the proportional income tax rate will, ceteris paribus, ...

A. reduce the size of the multiplier **B.** increase the marginal propensity to consume

C. leave the multiplier unchanged **D.** increase the size of the multiplier

Answer: A — Tax is a leakage; t appears in the denominator.

72. Does an increase in autonomous government spending change the value of the multiplier?

A. No, and it has no effect on equilibrium income

B. Yes, it increases the multiplier

C. Yes, it reduces the multiplier

D. No, it raises equilibrium income through the multiplier but does not change the multiplier itself

Answer: D — Only c, t and m determine alpha.

73. A higher marginal propensity to import ...

A. increases the multiplier because imports are an injection

B. increases autonomous spending

C. has no effect on the multiplier

D. reduces the multiplier because imports are a leakage

Answer: D — m enters the denominator.

74. If the multiplier is 2.5 and total autonomous spending is 800, equilibrium income is:

A. 2 000 **B.** 3 200

C. 320 **D.** 802.5

Answer: A — Y = alpha x A = 2.5 x 800.

75. If t = 0.2, equilibrium income is 2 000 and G = 450, the government budget shows a ...

A. deficit of 400 **B.** surplus of 400

C. deficit of 50 **D.** surplus of 50

Answer: C — T = 0.2 x 2 000 = 400; T - G = -50.

76. According to Keynes, ...

A. savings are always automatically re-injected as investment

B. supply creates its own demand (Say's law)

C. the economy is always at full employment

D. aggregate demand (spending) determines production and income

Answer: D — Keynes reversed Say's law: A drives Y.

77. Inflation is best defined as ...

A. an increase in the price of a single good

B. a decrease in the general price level

C. a continuous and considerable rise in prices in general

D. a once-off increase in the price of petrol

Answer: C — Key aspects: process, considerable, prices in general.

78. A once-off increase in the petrol price is ...

A. inflationary only if wages fall
B. always defined as hyperinflation
C. an example of deflation
D. not inflationary in itself, but its secondary effects on other prices are

Answer: D — The knock-on effects on other prices are inflationary.

79. The CPI was 120.0 in March 2025 and 124.2 in March 2026. The year-on-year inflation rate for March 2026 is:

A. 3.50%
B. 0.57%
C. 3.38%
D. 4.20%

Answer: A — (124.2 - 120.0) / 120.0 x 100 = 3.50%.

80. The CPI was 123.5 in February 2026 and 124.2 in March 2026. The month-on-month inflation rate for March 2026 is:

A. 0.57%
B. 3.50%
C. 0.56%
D. 0.70%

Answer: A — (124.2 - 123.5) / 123.5 x 100 = 0.57%.

81. If nominal GDP is R6 000 bn and real GDP is R5 000 bn, the GDP deflator is:

A. 120
B. 83.33
C. 20
D. 1.2

Answer: A — 6 000 / 5 000 x 100 = 120.

82. Cost-push inflation is illustrated by ...

A. a rightward shift of aggregate supply
B. a rightward shift of aggregate demand
C. a leftward shift of aggregate demand
D. a leftward shift of aggregate supply, raising prices and lowering output (stagflation)

Answer: D — Higher production costs reduce supply.

83. When the economy is at full employment, a further increase in aggregate demand will ...

A. raise output and lower prices
B. only raise real output
C. reduce unemployment significantly
D. only raise the general price level

Answer: D — AS is vertical at full employment.

84. In the structuralist approach, inflationary expectations and indexation are examples of ...

A. initiating factors
B. exogenous shocks
C. underlying factors
D. propagating factors

Answer: D — Propagating factors spread and sustain the inflation process.

85. The conflict approach sees inflation as ...

A. a symptom of disagreement between social groups over the distribution of income
B. caused solely by natural disasters
C. the result of falling productivity only
D. the result of too much money chasing too few goods only

Answer: A — Groups claim more than the available national income.

86. Who tends to gain from unexpected inflation?

A. Borrowers who repay fixed-interest loans
B. Lenders at fixed interest rates
C. Pensioners on fixed incomes
D. Savers who earn fixed interest

Answer: A — The real value of debt falls.

87. The original Phillips curve shows ...

A. the relationship between interest rates and money demand
B. the relationship between GDP and GNI
C. an inverse relationship between inflation and unemployment
D. a positive relationship between inflation and unemployment

Answer: C — Trade-off between inflation and unemployment; stagflation shifted it right.

88. Under the strict (narrow) definition, an unemployed person must also have ...

A. actively searched for work in the four weeks before the survey
B. registered for a UIF grant
C. worked for at least one hour in the week before the survey
D. been unemployed for longer than one year

Answer: A — The broad definition drops the active-search requirement.

89. Discouraged work-seekers are counted as unemployed ...

A. under the broad (expanded) definition only
B. under the strict (narrow) definition only
C. under both definitions
D. under neither definition

Answer: A — They want work but have stopped looking.

90. If 15 million people are employed and 5 million are unemployed (strict definition), the narrow unemployment rate is:

A. 25.00%
B. 75.00%
C. 20.00%
D. 33.33%

Answer: A — 5 / (15 + 5) x 100 = 25%.

91. Grape pickers in the Hex River Valley who are without work after the harvest are an example of ...

A. cyclical unemployment
B. frictional unemployment
C. seasonal unemployment
D. structural unemployment

Answer: C — Work required for only part of the year.

92. Bank tellers who lose their jobs because customers switch to app banking are an example of ...

A. structural unemployment
B. cyclical unemployment
C. frictional unemployment
D. seasonal unemployment

Answer: A — Technological change: skills mismatch.

93. Retail workers retrenched because sales fell during a recession are an example of ...

A. cyclical unemployment
B. frictional unemployment
C. structural unemployment
D. seasonal unemployment

Answer: A — Demand-deficient unemployment linked to the business cycle.

94. Which type of unemployment is temporary, always present in an economy and not regarded as a serious problem?

A. Structural unemployment
B. Cyclical unemployment
C. Frictional (search) unemployment
D. Long-term unemployment

Answer: C — Time taken to move between jobs.

95. Economic growth is best measured as the annual percentage change in ...

A. the money supply M3
B. the consumer price index
C. real GDP (preferably per capita)
D. nominal GDP

Answer: C — Must be in real terms and adjusted for population.

96. The phase of the business cycle between a peak and the next trough is the ...

A. boom
B. recovery
C. downswing (contraction)
D. upswing (expansion)

Answer: C — Falling incomes and rising unemployment.

97. Technically, a recession occurs when real GDP ...

A. grows slower than the population
B. grows by less than 3% per year
C. declines for two successive quarters
D. declines in any single month

Answer: C — Two successive quarters of negative growth.

98. Keynesian economists believe that business cycles are ...

A. random occurrences with no policy implications
B. exogenous, so government should not intervene
C. endogenous, so government should intervene with fiscal and monetary policy
D. caused by sunspots

Answer: C — Classical = exogenous; structuralist = random/structural shocks.

99. Which one of the following is a LEADING business cycle indicator?

A. Industrial production
B. Duration of unemployment
C. The exchange rate
D. Company profits already reported

Answer: C — Leading indicators predict future activity (oil price, exchange rates, gold price).

100. Capital deepening occurs when ...

A. the labour force grows faster than the capital stock
B. the capital stock grows at the same rate as the labour force
C. the amount of capital per worker increases
D. depreciation exceeds investment

Answer: C — Capital widening keeps capital per worker constant.

SECTION B / *AFDELING B* — THEORY — MEMORANDUM [150]

QUESTION 1 / *VRAAG 1* *Concepts and definitions* [10]

Define the following concepts:

1.1 Fiscal policy (1)

The use of government spending and taxation (the budget) to influence total spending/aggregate demand in the economy in order to achieve macroeconomic objectives such as growth, employment and price stability. (1)

1.2 Monetary policy (1)

Measures taken by the monetary authorities (SARB) to influence the quantity of money or the rate of interest with a view to achieving stable prices, full employment and economic growth. (1)

1.3 Unit of account (1)

The function of money as an agreed measure for stating the prices of goods and services, so that all values are expressed in money terms and can be compared. (1)

1.4 Comparative advantage (1)

The ability of a party (individual, firm or country) to produce a particular good or service at a lower opportunity cost than another party. (1)

1.5 Autonomous consumption (1)

Consumption that is independent of the level of income; the amount households consume even when income is zero, financed from past savings or credit (the intercept $\bar{C}$ of the consumption function). (1)

1.6 Gross domestic product (GDP) (1)

The total value of all final goods and services produced within the boundaries of a country in a particular period (usually one year). (1)

1.7 Inflation (1)

A continuous and considerable rise in prices in general (the general price level). (1)

1.8 Structural unemployment (1)

Unemployment confined to certain industries, sectors or categories of workers, caused by a mismatch between workers' qualifications/skills and job requirements, e.g. due to technological change or structural decline of industries. (1)

1.9 Capital widening (1)

When the stock of capital is increased to accommodate an increasing labour force, so that the amount of capital per worker stays the same. (1)

1.10 Gross domestic expenditure (GDE) (1)

The total value of spending within the borders of the country by households, firms and the government: GDE = C + I + G (includes imports, excludes exports). (1)

QUESTION 2 / *VRAAG 2* *Macroeconomic objectives* [8]

2.1 Name and briefly explain the five main macroeconomic objectives. (5)

- **Economic growth**: a sustained increase in the total (real) production of goods and services from one period to the next. (1)
- **Full employment**: all factors of production, particularly labour, are fully employed; unemployment kept as low as possible. (1)
- **Price stability**: keeping inflation low and stable – no hyperinflation or deflation. (1)
- **Balance of payments (external) stability**: balance between exports and imports; fairly stable BoP and exchange rate. (1)
- **Equitable distribution of income**: no excessive income disparities between members of the economy. (1)

2.2 Explain, with reference to the Phillips curve, why the objectives of full employment and price stability may conflict with each other. (3)

The original Phillips curve shows an inverse relationship (trade-off) between inflation and unemployment. (1) Policies that stimulate aggregate demand to reduce unemployment also push up the general price level (higher inflation), (1) while restrictive policies that lower inflation reduce spending, production and employment (higher unemployment). Policy makers therefore have to choose a combination, e.g. accept some inflation in return for lower unemployment. (1)

QUESTION 3 / *VRAAG 3* *National income accounting* [12]

3.1 In tabular form, distinguish between gross domestic product (GDP), gross domestic expenditure (GDE) and gross national income (GNI). Your table should show what each measures and the formula used to calculate it. (6)

	GDP	**GDE**	**GNI**
What it measures (1½)	Total value of final goods and services produced within the borders of the country (½)	Total spending within the borders of the country by residents (½)	Total income earned by the permanent residents of a country, wherever they are (½)
Formula (1½)	C + I + G + (X − Z) (½)	C + I + G (½)	GDP + primary income receipts − primary income payments to the rest of the world (½)
Key feature (3)	Includes production by foreigners in SA; best measure of economic activity (1)	Includes imports, excludes exports (1)	Includes income of SA residents abroad, excludes foreigners' income in SA; better measure of standard of living (1)

3.2 Indicate whether each of the following would be included in South Africa's GDP for 2026. Give a reason for each answer. (4)
(a) A Toyota bakkie manufactured in 2021 and resold in 2026.
(b) The salary of a Zimbabwean engineer working temporarily in Sandton.
(c) Vegetables grown by a household in its own garden for its own use.
(d) The output of a South African-owned clothing factory in Maseru, Lesotho.

(a) Not included – resale of a second-hand good; it was counted when first produced (no current production). (1)
(b) Included – production within SA's borders counts regardless of who produces it. (1)
(c) Not included – not sold in a formal market (non-market production is generally excluded). (1)
(d) Not included in SA's GDP – produced outside SA's borders (part of Lesotho's GDP; it would be in SA's GNI). (1)

3.3 Explain why GDP is called a *gross* measure and give two reasons why GDP is not a perfect measure of economic welfare. (2)

Gross: it does not subtract depreciation (consumption of fixed capital), which is difficult to calculate accurately. (1) Not a perfect welfare measure (any two, ½ each): ignores non-market production and the informal/unrecorded economy; does not account for negative externalities such as pollution ("bads"); says nothing about the distribution of income; data are provisional and revised. (1)

QUESTION 4 / *VRAAG 4* *Money and banking* [24]

4.1 Name and explain the three functions of money. Give an example of each. (3)

Medium of exchange – generally accepted as payment for goods and services or settlement of debt, e.g. paying for groceries with a card. (1)
Unit of account – an agreed measure for stating prices, e.g. a loaf of bread priced at R19.99. (1)
Store of value – the most liquid way to keep wealth; can be saved and used later, e.g. keeping money in a savings account. (1)

4.2 "Money is not the same as income or wealth." Explain this statement. (2)

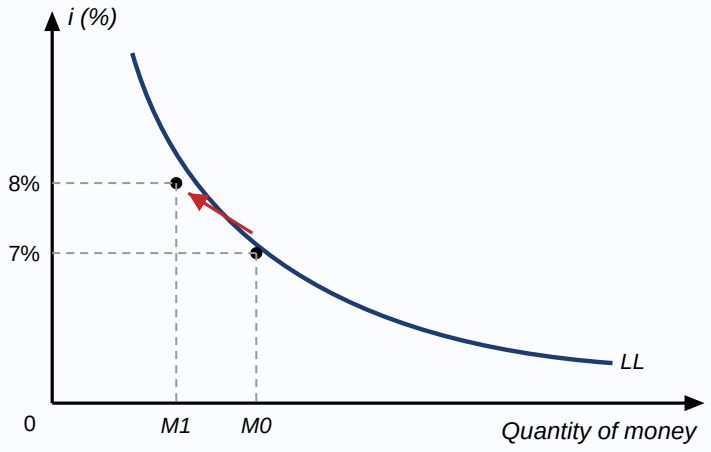
Income is the reward earned in the production process (wages, rent, interest, profit) – a flow over a period. (1) Wealth consists of assets accumulated over time (property, shares, paintings). Both are measured in money terms, but many wealthy people hold little money, especially during inflation when money loses its store-of-value function. (1)

4.3 Discuss the three motives for holding money (the demand for money). In your answer, distinguish between active and passive balances and state the factors that determine each. (6)

Transactions motive – money held to pay for day-to-day expenses (medium of exchange). (1)
Precautionary motive – money held for unplanned/unexpected expenditure. (1)
Speculative motive – money held as a store of value while waiting for better investment opportunities (e.g. bonds). (1)

Active balances (L1) = transactions + precautionary demand; mainly determined by the level of income – higher income → more money demanded (L1 shifts right). (1½)
Passive balances (L2) = speculative demand; determined by the interest rate (and expectations about it) – inverse relationship. (1)
Total demand LL = L1 + L2, so L = f(Y, i). (½)

4.4 With the aid of a clearly labelled graph, illustrate the effect on the quantity of money demanded in South Africa if interest rates increase from 7% to 8%. Also state how the graph would change if national income increased instead. (4)



Interest rate on the vertical axis, quantity of money on the horizontal axis, downward-sloping LL (1). The rise in i from 7% to 8% causes a **movement along** LL from M0 to M1 – a smaller quantity of money is demanded (1½), because the opportunity cost of holding money (interest forgone) rises (½). An increase in income would **shift LL to the right** (more active balances demanded at every interest rate). (1)

4.5 Explain how banks create money. Distinguish between passive and active money creation, and explain why there is no independent money supply curve in South Africa. (5)

Banks create demand deposits by making loans to creditworthy borrowers in response to the demand for loans. (1) **Passive** creation: a customer deposits cash and receives an equal demand deposit – the composition of money changes but the money supply (M = C + D) is unchanged. (1) **Active** creation: the bank grants credit (e.g. an overdraft) and creates a deposit without a cash deposit – the money supply increases. (1) Money creation is limited by cash-reserve and liquidity requirements, and the SARB regulates it by influencing the price of loans (the interest rate) via the repo rate. (1) The quantity of money is therefore determined by the interaction between the demand for money and the interest rate – it is demand-determined (endogenous), so there is no independent money supply curve. (1)

4.6 Name and briefly explain four functions of the South African Reserve Bank. (4)

Any four (1 each): formulation and implementation of monetary policy (repo rate); service to the government (banker and adviser; custodian of gold and foreign reserves; administering exchange control); provision of economic and statistical services (e.g. Quarterly Bulletin); maintaining financial stability (bank supervision, overseeing the national payment system); banker to other banks / lender of last resort; issuing (and destroying) banknotes and coins.

QUESTION 5 / *VRAAG 5* *Market failure and government intervention* **[26]**

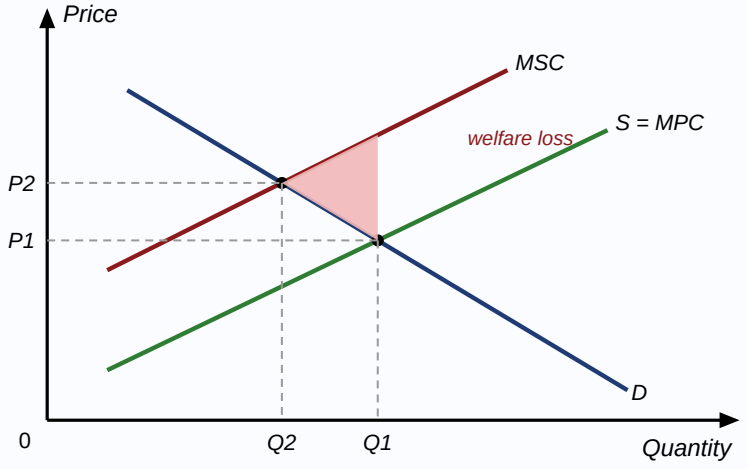
5.1 Name and briefly explain four main causes (types) of market failure. (4)

Monopoly and imperfect competition – a firm restricts output and charges high prices (1); public goods – non-rival and non-excludable, so markets under-provide them (1); externalities – costs/benefits borne by third parties not involved in the transaction (1); common property resources – rival but non-excludable resources that are over-used (tragedy of the commons) (1).

5.2 Classify each of the following according to rivalry and excludability, and name the type of good: (a) national defence; (b) a ticket to a movie at a cinema; (c) perlemoen (abalone) along the Cape coast; (d) a toll road that is not congested. (4)

Good	Rival?	Excludable?	Type of good
(a) National defence	No	No	Pure public good (1)
(b) Cinema ticket	Yes	Yes	Private good (1)
(c) Perlemoen	Yes	No	Common property resource (1)
(d) Uncongested toll road	No	Yes	Club/toll good (1)

5.3 With the aid of a clearly labelled graph, explain why a chemical factory that discharges waste into a river leads to an inefficient allocation of resources. (6)



Correct axes, D, S = MPC and MSC above MPC (2). The pollution imposes an external cost on third parties (water treatment, fishermen, recreation), so marginal social cost = MPC + external cost > MPC (1). The market produces where D = MPC at Q1 and P1 (½), but the socially efficient output is where D = MSC at Q2 and P2 (½). Q1 > Q2: the good is over-produced and under-priced (1); the shaded area is the welfare loss to society – a market failure (1).

5.4 Name two ways in which the government can reach the socially efficient output in Question 5.3. (2)

Levy a tax on the polluter equal to the difference between MSC and MPC to internalise the external cost (1); direct controls/regulation (e.g. pollution limits, permits) (1). (Coase bargaining is possible only if parties can bargain – rarely in practice.)

5.5 Governments, like markets, can fail. Explain government failure with reference to the behaviour of politicians and bureaucrats, and rent seeking. (4)

Government consists of people with their own motives. (½) **Politicians** aim to maximise votes and stay in office; they favour popular, short-term quick fixes (policy myopia) and are reluctant to admit mistakes. (1½) **Bureaucrats** maximise their status, salary and power, push for bigger budgets and staff, face no market competition and are less cost-conscious. (1) **Rent seeking**: firms, households and interest groups try to benefit at the expense of society through the political process (e.g. lobbying for protection), diverting resources to special interests. (1)

5.6 Briefly distinguish between the allocation, distribution and stabilisation functions of government. (3)

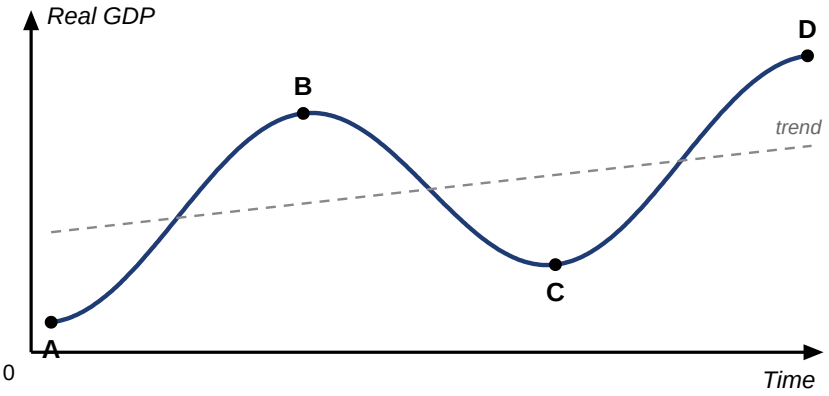
Allocation – providing goods and services the market under-supplies (roads, education, health). (1) Distribution – redistributing income and wealth more fairly (progressive taxes, social grants). (1) Stabilisation – reducing fluctuations in output, employment and prices (fiscal policy over the cycle). (1)

5.7 Discuss the three criteria of a good tax. (3)

Neutrality – minimal distortion of prices and economic decisions. (1) Equity – a fair burden: horizontal equity (same position, same tax), vertical equity (different positions, different tax / ability to pay) and the benefit principle. (1) Administrative simplicity – easy to understand and comply with, cheap to administer and hard to evade. (1)

QUESTION 6 / *VRAAG 6* *Economic growth and business cycles* [12]

Study the diagram below and answer the questions that follow.



6.1 Name the turning points B and C and explain the economic activity that takes place between points A and B, and between points B and C. (4)

B = peak (½); C = trough (½). A to B: upswing/expansion (recovery and boom) – production, income and spending increase and unemployment declines. (1½) B to C:

downswing/contraction (recession) – falling production and incomes, rising unemployment; two successive quarters of negative growth = recession. (1½)

6.2 Classify each statement according to the theory of business cycles (classical, Keynesian or structuralist): (3)
(a) "The economy is inherently stable; cycles are caused by external factors such as droughts and wars, so government should leave the market alone."
(b) "Cycles are part of the way a market economy works; government should use fiscal and monetary policy to smooth peaks and troughs."
(c) "The market is neither inherently stable nor unstable; cycles result from structural changes and unpredictable events such as the 2008/9 financial crisis."

(a) Classical (exogenous) (1); (b) Keynesian (endogenous) (1); (c) Structuralist (random/structural events) (1).

6.3 Classify each of the following as a leading, coincident or lagging business-cycle indicator: (3)
(a) the rand–dollar exchange rate; (b) the volume of industrial production; (c) the average duration of unemployment.

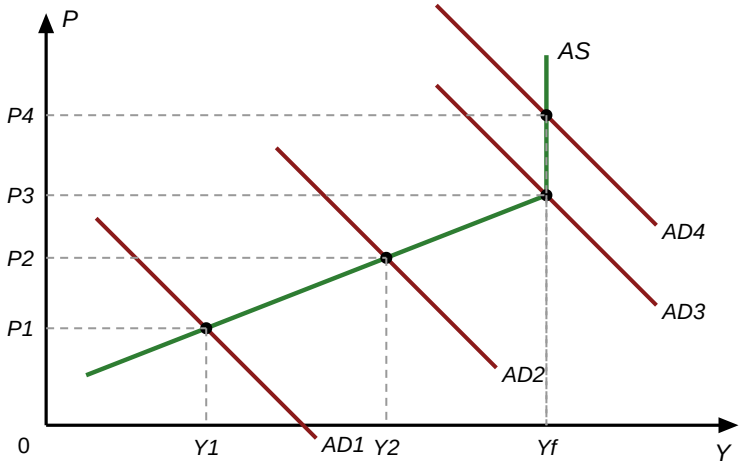
(a) Leading (1); (b) Coincident (1); (c) Lagging (1).

6.4 Name two supply-side and two demand-side sources of economic growth. (2)

Supply (any two, ½ each): natural resources, labour (quantity and quality / human capital), capital (incl. technology), entrepreneurship. Demand (any two, ½ each): domestic demand (C, I, G), export demand (X), import substitution.

QUESTION 7 / *VRAAG 7* *Inflation* [18]

Study the diagram below.

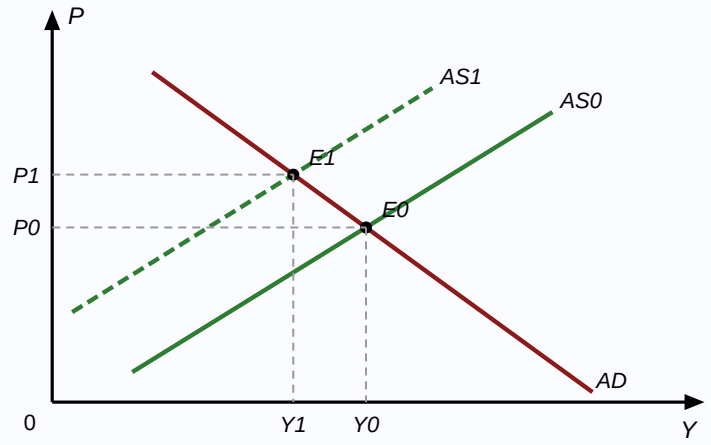


7.1 Identify the type of inflation illustrated in the diagram and name any three factors that can cause this type of inflation. What happens when the economy moves from E3 to E4? (4)

Demand-pull inflation (1). Causes (any three, ½ each): increase in C (more credit, higher wages), increase in I (lower interest rates, better profit expectations), increase in G, increase in export earnings (X), increase in the money stock. (1½) From E3 to E4 the economy is at full

employment (Yf), so the increase in AD only raises the price level from P3 to P4 – output does not increase. (1½)

7.2 With the aid of a graph, explain cost-push inflation. Explain why restrictive monetary and fiscal policies are not appropriate to combat this type of inflation, and name the policy that is preferred. (6)



Graph: AS shifts left from AS0 to AS1 with AD unchanged (1½). Prices are pushed up by increases in production costs (wages, imported inputs, profit margins, lower productivity, natural disasters) (1); the price level rises from P0 to P1 while output falls from Y0 to Y1 – stagflation (1). Restrictive monetary/fiscal policy would lower AD further and deepen the recession and unemployment (1½). Preferred: an incomes policy (agreement between government, employers and unions to limit wage and price increases) (1).

7.3 Explain the three sets of factors identified by the structuralist approach to inflation. (3)

Underlying factors – the background that makes the economy vulnerable to inflation, e.g. lack of fiscal discipline, size of the public sector. (1) Initiating factors – trigger or intensify the inflation process (demand-pull, cost-push and other price shocks, e.g. a drought). (1) Propagating factors – spread and sustain the shock through the economy, e.g. inflationary expectations, indexation, wage-price spirals. (1)

7.4 Briefly explain the conflict approach to inflation. (2)

Inflation is a symptom of disharmony between social groups (unions, firms, government) about the distribution of income. (1) Each group tries to claim a bigger share of national income than is available, which pushes up wages and prices; inflation reflects the lack of mechanisms to reconcile these claims. (1)

7.5 In tabular form, provide three differences between the consumer price index (CPI) and the producer price index (PPI). (3)

CPI	PPI
Measures price changes facing households/consumers (cost of living) (½)	Measures cost pressures facing producers (½)
Basket of consumer goods and services (½)	Basket of capital and intermediate goods (½)
Retail/final prices including VAT (½)	Wholesale/production prices, excluding VAT (½)

QUESTION 8 / *VRAAG 8* *International trade and exchange rates* [16]

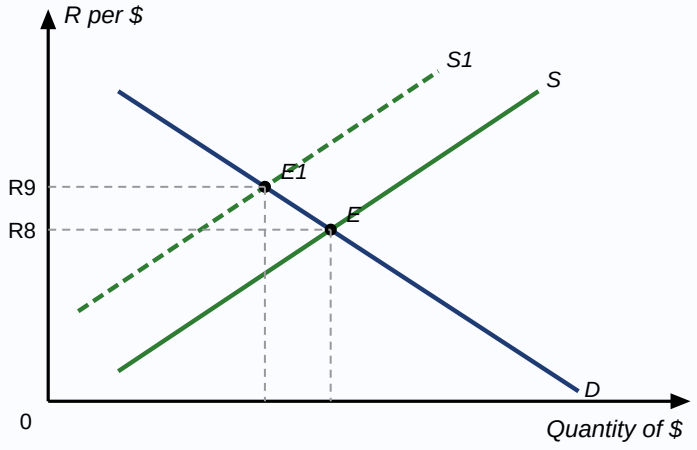
8.1 Define globalisation and give two pieces of evidence of globalisation. (2)

Globalisation is the process of deeper economic integration between countries and regions of the world. (1) Evidence (any two, ½ each): expanded trade, increased (integrated) capital markets, more tourism, new technology improving links, highly mobile factors of production, rapid spread of shocks (e.g. 2008/9 crisis).

8.2 Indicate whether each transaction represents a demand for or a supply of US dollars in the South African foreign exchange market: (4)
(a) A South African student pays her tuition at a US university.
(b) A US investor buys Sasol shares on the JSE.
(c) A South African mine sells platinum to an American car maker.
(d) A South African investor expects the rand to depreciate against the dollar.

(a) Demand for dollars (1); (b) Supply of dollars (1); (c) Supply of dollars (1); (d) Demand for dollars (1).

8.3 Use a graph to illustrate the effect of a decrease in the supply of dollars on the rand–dollar exchange rate (initial rate $1 = R8). Has the rand appreciated or depreciated? What is the effect on South African exports and imports? (4)



Graph: S shifts left to S1, new equilibrium E1 at a higher rand price of dollars (e.g. $1 = R9) (2). The rand has depreciated (dollar appreciated) (1). SA exports become cheaper for

foreigners (exports rise) and imports become more expensive (imports fall) – the current account tends to improve (1).

8.4 List two possible reasons for an appreciation of the rand against the US dollar. (2)

Any two (1 each): higher prices of SA export commodities (gold, platinum); increase in SA exports; increased foreign investment in SA (portfolio or direct); higher SA interest rates relative to the USA; expectations that the rand will appreciate; increased tourism to SA; decrease in SA imports.

8.5 Discuss two arguments in favour of and two arguments against the use of trade barriers. (4)

For (any two, 1 each): improve the balance of payments; protect against dumping; neutralise foreign export subsidies; protect infant industries; protect employment; raise government revenue; national security. Against (any two, 1 each): retaliation and trade wars; welfare cost – higher prices and less choice for consumers; inefficiency – protected firms have little incentive to innovate.

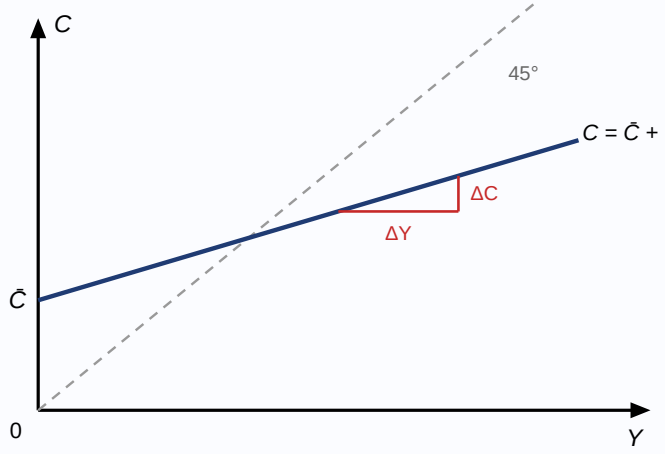
QUESTION 9 / *VRAAG 9* *The Keynesian model* [12]

9.1 Provide four assumptions of the simple Keynesian model. (2)

Any four (½ each): only households and firms (no G, X, Z) initially; prices and wages are given/constant; the money stock and interest rates are given (exogenous); spending is the driving force in the economy (demand determines production); there are unemployed resources; investment is autonomous.

9.2 What is meant by the consumption function? Explain its three main characteristics and illustrate the function graphically. (5)

The consumption function is the relationship between consumption expenditure by households and total income: $C = \bar{C} + cY$. (1) (i) Consumption increases as income increases (positive relationship). (½) (ii) Consumption is positive even when income is zero (autonomous consumption, financed by past savings or credit). (½) (iii) When income increases, consumption increases by less than the increase in income – $0 < c < 1$; the rest is saved (slope = MPC = $\Delta C/\Delta Y$). (1)



Graph: positive intercept $\bar{C}$, upward slope flatter than 45° (2).

9.3 Name three factors that determine investment spending by firms. (3)

Any three (1 each): the interest rate (inverse relationship – cost of borrowing); the cost of capital goods; expected revenue/profit from capital goods; business confidence and willingness to take risks / expected future economic conditions.

9.4 Explain how (a) an increase in the tax rate and (b) an increase in government spending affect the multiplier. (2)

(a) A higher tax rate increases leakages out of the circular flow, increasing the denominator $1 - c(1 - t) + m$, so the multiplier becomes smaller. (1) (b) Government spending is autonomous – it does not change the multiplier; it increases equilibrium income by $\alpha \times \Delta G$. (1)

QUESTION 10 / *VRAAG 10* *Unemployment* [12]

10.1 Identify the type of unemployment in each case and justify your answer: (4)
(a) Lerato, a new BCom graduate, is searching for her first job.
(b) Mineworkers lose their jobs because a mine introduces automated drilling machines.
(c) Waiters at a Durban beach resort have no work after the December holidays.
(d) Workers at a furniture factory are laid off because sales fell during a recession.

(a) Frictional – time taken by first-time job seekers to find a job (1). (b) Structural – change in production methods (mechanisation) creates a skills mismatch (1). (c) Seasonal – tourism jobs required only for part of the year (1). (d) Cyclical (demand-deficient) – fall in total demand during a downswing (1).

10.2 Distinguish between the strict (narrow) and broad (expanded) definitions of unemployment. (2)

Both: persons aged 15+ (15–64) who did not work in the reference week and were available to work. (½) Strict: must also have actively searched for work (or tried to start a business) in the four weeks before the survey. (½) Broad: drops the active-search requirement – only a desire to work is needed, so it includes discouraged work-seekers. (1)

10.3 Give two reasons why people enter the unemployment pool and two reasons why they leave it. (2)

Enter (½ each): new entrants to the labour market; leaving a previous job to look for another; being retrenched/laid off; being fired. Leave (½ each): finding a new job/being hired; being recalled after a lay-off; becoming discouraged and stopping the search (leaving the labour force).

10.4 Name three possible causes of structural unemployment in South Africa and one policy aimed at the supply of labour that could reduce it. (4)

Causes (any three, 1 each): lack of the required education, training or skills; changes in production methods (mechanisation/automation); changes in the goods and services demanded; foreign competition (e.g. dumping); structural decline of certain industries; discrimination. Policy (1): improve education, training and skills / retraining of workers (or limit population growth / stricter immigration control).

SECTION C / *AFDELING C* — CALCULATIONS — MEMORANDUM [150]

QUESTION 1 / *VRAAG 1* *GDP and related aggregates* [20]

Table 1 provides key national accounts data for a hypothetical South African economy in 2025. Use it to answer the questions that follow. Show all formulae and calculations.

Item	R million
Household consumption expenditure	2 480 000
Gross capital formation (investment)	620 000
Government consumption expenditure	760 000
Exports of goods and services	1 050 000
Imports of goods and services	1 010 000
Taxes on production and imports	455 000
Subsidies on production	38 000
Consumption of fixed capital (depreciation)	410 000
Salaries earned by South African engineers working in Dubai	96 000
Dividends paid to British shareholders of South African mining companies	214 000
Population (persons)	63 100 000

1.1 Calculate gross domestic expenditure (GDE) at market prices. (2)

> GDE = C + I + G (½)
> = 2 480 000 + 620 000 + 760 000 (1)
> = **R3 860 000 million** (½)

1.2 Calculate GDP at market prices using the expenditure method. (2)

> GDP@MP = C + I + G + X − Z (½)
> = 3 860 000 + 1 050 000 − 1 010 000 (1)
> = **R3 900 000 million** (½)

1.3 Calculate GDP at factor cost. (2)

> GDP@FC = GDP@MP − taxes on production and imports + subsidies (½)
> = 3 900 000 − 455 000 + 38 000 (1)
> = **R3 483 000 million** (½)

1.4 Calculate gross national income (GNI) at market prices. (3)

> GNI@MP = GDP@MP + primary income receipts − primary income payments (1)
> Receipts = salaries of SA engineers in Dubai = 96 000; payments = dividends to British

shareholders = 214 000 (1)
GNI@MP = 3 900 000 + 96 000 − 214 000 = **R3 782 000 million** (1)

1.5 Calculate net national income (national income) at factor cost. (3)

NNI@FC = GNI@MP − depreciation − taxes + subsidies (1)
= 3 782 000 − 410 000 − 455 000 + 38 000 (1)
= **R2 955 000 million** (1)

1.6 Calculate net domestic product (NDP) at factor cost. (2)

NDP@FC = GDP@FC − depreciation (½)
= 3 483 000 − 410 000 (1)
= **R3 073 000 million** (½)

1.7 Calculate GDP per capita (at market prices) in rand. Round your answer to two decimal places. (2)

GDP per capita = GDP@MP ÷ population (½)
= R3 900 000 000 000 ÷ 63 100 000 (1)
= **R61 806.66** per person (½)

1.8 Explain why GNI is smaller than GDP in this economy, and what the difference between GDP and GDE tells you about trade with the rest of the world. (4)

GNI < GDP because primary income payments to foreigners (R214 000 m) exceed primary income received from abroad (R96 000 m): net primary income payments of R118 000 m flow out (2). GDP − GDE = X − Z = R40 000 m > 0: exports exceed imports, so the country has a trade surplus – production exceeds domestic spending (2).

QUESTION 2 / *VRAAG 2* *GDP: the production and income methods* [15]

On 12 December 2025, Vaal Mining Ltd hosts its year-end gala and pays the event planner, **Mosaic Events**, R2 450 000 to organise it. Mosaic Events rents a hall for R180 000, hires 20 waiters at R4 500 each, and pays R25 000 interest on a business loan. Mosaic Events contracts **Chef Lerato** to cater for the evening at a fee of R1 360 000. Chef Lerato rents a commercial kitchen for R75 000 and pays wages of R140 000. She buys all her ingredients from **FreshMart Wholesalers** for R640 000. FreshMart pays R48 000 rent and R112 000 wages, and buys its fresh produce from **Lekoa Farms** for R230 000. Lekoa Farms pays wages of R86 000 and R14 000 interest on a tractor loan. Lekoa Farms does not buy any intermediate goods. (Hint: profit is the balancing item.)

2.1 Complete the table in the **Annexure** and show the value of GDP according to the production (value-added) method. (6)

Participant	Sales	Purchases	Value added	Wages	Rent	Interest	Profit	Income
Lekoa Farms	230 000	0	230 000	86 000	–	14 000	130 000	230 000
FreshMart	640 000	230 000	410 000	112 000	48 000	–	250 000	410 000
Chef Lerato	1 360 000	640 000	720 000	140 000	75 000	–	505 000	720 000
Mosaic Events	2 450 000	1 360 000	1 090 000	90 000	180 000	25 000	795 000	1 090 000
TOTAL	4 680 000	2 230 000	**2 450 000**	428 000	303 000	39 000	1 680 000	**2 450 000**

Value added = sales − intermediate purchases for each participant (½ per correct value-added figure = 2); correct purchases column (2); GDP (production method) = **R2 450 000** (2).

2.2 Use the table in the Annexure to show the value of GDP according to the income method. Show how you calculated the profit of each participant. (6)

> Profit = value added − wages − rent − interest
> Lekoa: 230 000 − 86 000 − 14 000 = 130 000 (1)
> FreshMart: 410 000 − 112 000 − 48 000 = 250 000 (1)
> Chef Lerato: 720 000 − 140 000 − 75 000 = 505 000 (1)
> Mosaic: 1 090 000 − 90 000 − 180 000 − 25 000 = 795 000 (1)
> GDP (income) = wages 428 000 + rent 303 000 + interest 39 000 + profit 1 680 000 = **R2 450 000** (2)

2.3 State GDP according to the expenditure method and explain why adding up the value of all sales would give a misleading figure. (3)

Expenditure method: only the final sale counts – the R2 450 000 paid by Vaal Mining (1). Sum of all sales = R4 680 000 (½), which double counts intermediate goods (ingredients, catering) already included in the final price, overstating GDP by R2 230 000 (1½).

QUESTION 3 / *VRAAG 3* *Nominal GDP, real GDP and the GDP deflator* [15]

A small economy produces only three goods. Use the information below to answer the questions. **2024 is the base year.**

Good	2024 quantity	2024 price	2025 quantity	2025 price
Bread (loaves)	1 200	R18.00	1 300	R19.80
Airtime (vouchers)	800	R25.00	850	R26.50
Taxi trips	1 500	R14.00	1 560	R15.40

3.1 Calculate nominal GDP for 2024 and 2025. (4)

Nominal GDP = Σ (current P × current Q) (½)
2024: (1 200 × 18) + (800 × 25) + (1 500 × 14) = 21 600 + 20 000 + 21 000 = **R62 600** (1½)
2025: (1 300 × 19.80) + (850 × 26.50) + (1 560 × 15.40) = 25 740 + 22 525 + 24 024 = **R72 289** (2)

3.2 Calculate real GDP for 2024 and 2025. (3)

Real GDP = Σ (base-year P × current Q) (½)
2024: = nominal GDP in base year = **R62 600** (½)
2025: (1 300 × 18) + (850 × 25) + (1 560 × 14) = 23 400 + 21 250 + 21 840 = **R66 490** (2)

3.3 Calculate the growth rate of nominal GDP and of real GDP between 2024 and 2025. (4)
Which one measures economic growth? Explain.

Nominal growth = (72 289 − 62 600) ÷ 62 600 × 100 = **15.48%** (1½)
Real growth = (66 490 − 62 600) ÷ 62 600 × 100 = **6.21%** (1½)
Real growth (6.21%) measures economic growth – it removes the effect of price increases; nominal growth mixes growth and inflation (1)

3.4 Calculate the GDP deflator for 2024 and 2025, and the inflation rate for 2025. (4)

Deflator = nominal GDP ÷ real GDP × 100 (½)
2024: 62 600 ÷ 62 600 × 100 = **100.00** (½)
2025: 72 289 ÷ 66 490 × 100 = **108.72** (1)
Inflation 2025 = (108.72 − 100) ÷ 100 × 100 = **8.72%** (2)

QUESTION 4 / *VRAAG 4* *Measuring inflation with the CPI* [12]

The table shows hypothetical consumer price index (CPI) values (December 2024 = 100).

Month	CPI 2025	CPI 2026
January	104.2	107.8
February	104.9	108.6
March	105.6	109.3
April	105.9	110.1

4.1 Calculate the year-on-year inflation rate for each month of 2026. (4)

Y-o-y inflation = (CPI month 2026 − CPI same month 2025) ÷ CPI same month 2025 × 100
January: (107.8 − 104.2) ÷ 104.2 × 100 = **3.45%** (1)
February: (108.6 − 104.9) ÷ 104.9 × 100 = **3.53%** (1)
March: (109.3 − 105.6) ÷ 105.6 × 100 = **3.50%** (1)
April: (110.1 − 105.9) ÷ 105.9 × 100 = **3.97%** (1)

4.2 Calculate the month-on-month inflation rate for February, March and April 2026. (3)

M-o-m inflation = (CPI this month − CPI previous month) ÷ CPI previous month × 100
February: (108.6 − 107.8) ÷ 107.8 × 100 = **0.74%** (1)
March: (109.3 − 108.6) ÷ 108.6 × 100 = **0.64%** (1)
April: (110.1 − 109.3) ÷ 109.3 × 100 = **0.73%** (1)

4.3 Using the inflation target range in your study material (3–6%), comment on whether the SARB is meeting its target in April 2026, and what the MPC would likely do if inflation rose above 6%. (2)

April 2026 y-o-y inflation of 3.97% lies within the 3–6% target range, so the target is being met (1). If inflation rose above 6%, the MPC would likely raise the repo rate (contractionary monetary policy) to curb spending (1).

4.4 Explain why year-on-year rates are preferred to month-on-month rates when reporting inflation. (3)

Month-on-month changes are small and volatile and are affected by seasonal factors (e.g. December spending, annual tariff increases in April) (1½). Comparing the same month a year apart removes seasonal effects and expresses inflation as an annual rate, making it comparable with the target and between years (1½).

QUESTION 5 / *VRAAG 5* *The labour market* [15]

Labour market statistics for the fictional country of Mzansia for the third quarter of 2026 are as follows:

Category	Number (thousands)
Employed	16 800
Unemployed (actively searched for work in the past four weeks)	8 200
Not economically active:	
Discouraged work-seekers	3 400
Full-time students	6 900
Homemakers	2 700
Ill or disabled	1 500
Other (early retirees, etc.)	1 700
Working-age population (15–64)	**41 200**

5.1 Calculate the labour force (economically active population) according to the strict definition. (2)

Labour force = employed + unemployed (½)
= 16 800 + 8 200 (1)
= **25 000 thousand (25 million)** (½)

5.2 Calculate the narrow (official) unemployment rate. (3)

Narrow UR = unemployed ÷ labour force × 100 (1)
= 8 200 ÷ 25 000 × 100 (1)
= **32.80%** (1)

5.3 Calculate the broad (expanded) unemployment rate. (4)

Broad unemployed = 8 200 + 3 400 = 11 600 (1)
Broad labour force = 25 000 + 3 400 = 28 400 (1)
Broad UR = 11 600 ÷ 28 400 × 100 (1)
= **40.85%** (1)

5.4 Calculate the labour force participation rate (strict definition). (3)

LFPR = labour force ÷ working-age population × 100 (1)
= 25 000 ÷ 41 200 × 100 (1)
= **60.68%** (1)

5.5 Explain why the broad unemployment rate is higher than the narrow rate, and give one argument for using the broad definition in South Africa. (3)

The broad definition adds discouraged work-seekers (who want work but have stopped searching) to both the number of unemployed and the labour force; since they are all unemployed, the rate rises (1½). Argument: in SA many people stop searching because jobs are scarce in their area or search is too costly (transport), so the narrow rate understates the true extent of joblessness (1½).

QUESTION 6 / *VRAAG 6* *Money supply* [9]

Use the following hypothetical data for South Africa to answer the questions. Show all equations.

Item	R billion
Coins and banknotes in circulation outside the banking sector	140
Banknotes and coins held in bank vaults	30
Demand deposits (cheque and transmission accounts) of the domestic private sector	1 260
Short- and medium-term deposits of the domestic private sector	1 480
Long-term deposits of the domestic private sector	1 320
Deposits of the central government with banks	95

6.1 Calculate M1. (2)

M1 = coins and notes in circulation + demand deposits (½)
= 140 + 1 260 (1)
= **R1 400 billion** (½)

6.2 Calculate M2. (2)

M2 = M1 + short- and medium-term deposits (½)
= 1 400 + 1 480 (1)
= **R2 880 billion** (½)

6.3 Calculate M3. (2)

M3 = M2 + long-term deposits (½)
= 2 880 + 1 320 (1)
= **R4 200 billion** (½)

6.4 Identify the items you excluded and explain why. (2)

Notes and coins in bank vaults (R30 bn) – not in circulation; they are banks' cash reserves (1). Central government deposits (R95 bn) – money supply measures include only deposits of the domestic private sector (1).

6.5 Which measure of money corresponds most closely to money's function as a store of value? Explain. (1)

M3 – it includes long-term deposits (longer than six months), which are held as a store of value rather than for spending. (1)

QUESTION 7 / *VRAAG 7* *Taxation* [16]

The table below shows the SARS individual income tax rates for the 2024/25 tax year.

Taxable income (R)	**Rates of tax (R)**
1 – 237 100	18% of taxable income
237 101 – 370 500	42 678 + 26% of taxable income above 237 100
370 501 – 512 800	77 362 + 31% of taxable income above 370 500
512 801 – 673 000	121 475 + 36% of taxable income above 512 800
673 001 – 857 900	179 147 + 39% of taxable income above 673 000
857 901 – 1 817 000	251 258 + 41% of taxable income above 857 900
1 817 001 and above	644 489 + 45% of taxable income above 1 817 000
Rebates	Primary: R17 235 | Secondary (65 and older): R9 444 | Tertiary (75 and older): R3 145

7.1 Naledi (42 years old) earns a monthly salary of R58 500. Calculate the total annual tax payable by Naledi. (4)

Annual income = 58 500 × 12 = R702 000 (1)
Tax = 179 147 + 0.39 × (702 000 − 673 000) − 17 235 (1)
= 179 147 + 11 310 − 17 235 (1)
= **R173 222** (1)

7.2 Calculate Naledi's effective (average) tax rate. (2)

> ETR = tax payable ÷ taxable income × 100 (½)
> = 173 222 ÷ 702 000 × 100 (1)
> = **24.68%** (½)

7.3 What is Naledi's marginal tax rate? Explain what it means. (1)

> **39%** (½) – each additional rand she earns is taxed at 39 cents (½).

7.4 Naledi's father, aged 68, has a taxable income of R310 000. Calculate his tax payable and his effective tax rate. (4)

> Tax = 42 678 + 0.26 × (310 000 − 237 100) − 17 235 − 9 444 (1½)
> = 42 678 + 18 954 − 26 679 = **R34 953** (1)
> ETR = 34 953 ÷ 310 000 × 100 = **11.28%** (1½)

7.5 Sipho earns R12 000 per month and spends all of it on goods subject to VAT. Thandeka earns R80 000 per month and spends 60% of her income on goods subject to VAT. Assume VAT of 15% is levied on the amount spent. Calculate the average VAT rate (VAT paid as a percentage of income) for each person and use your answer to classify VAT. (5)

> Sipho: VAT = 0.15 × 12 000 = R1 800; average rate = 1 800 ÷ 12 000 × 100 = **15%** (2)
> Thandeka: spending = 0.6 × 80 000 = R48 000; VAT = 0.15 × 48 000 = R7 200; average rate = 7 200 ÷ 80 000 × 100 = **9%** (2)
> VAT is **regressive** – it takes a bigger share of income from the lower-income earner (1)

QUESTION 8 / *VRAAG 8* *International trade and exchange rates* [18]

South Africa and Brazil are the only two countries in the world, producing only citrus and coffee. The table shows the output of one worker per month if all of the worker's time is devoted to one good.

Country	Citrus (tonnes)	Coffee (bags)
South Africa	60	20
Brazil	80	40

8.1 Which country has an absolute advantage in the production of each good? Explain. (2)

> Brazil has an absolute advantage in both citrus (80 > 60) and coffee (40 > 20) (1), because with the same resources it produces more of both goods (1).

8.2 Calculate the opportunity cost of producing 1 tonne of citrus and 1 bag of coffee in each country. (4)

> 1 tonne citrus: SA = 20 ÷ 60 = **0.33 bags of coffee** (1); Brazil = 40 ÷ 80 = **0.50 bags** (1)
> 1 bag coffee: SA = 60 ÷ 20 = **3 tonnes of citrus** (1); Brazil = 80 ÷ 40 = **2 tonnes** (1)

8.3 Which country has a comparative advantage in each good? Which good should each country specialise in and export? (2)

South Africa has a comparative advantage in citrus (lower opportunity cost 0.33 < 0.50) and should export citrus (1). Brazil has a comparative advantage in coffee (2 < 3) and should export coffee (1).

8.4 Between which values must the international terms of trade for 1 bag of coffee lie for trade to be mutually beneficial? (2)

2 tonnes of citrus < 1 bag of coffee < 3 tonnes of citrus (2). Brazil exports coffee only if it receives more than 2 tonnes; SA imports coffee only if it pays less than 3 tonnes.

8.5 Suppose the countries agree to trade at 1 bag of coffee = 2.5 tonnes of citrus. Show how both countries gain. (2)

SA obtains a bag of coffee for 2.5 tonnes instead of the 3 tonnes it would sacrifice producing it itself – a gain of 0.5 tonnes per bag (1). Brazil receives 2.5 tonnes of citrus per bag instead of 2 tonnes at home – a gain of 0.5 tonnes per bag (1).

8.6 South Africa's export price index is 128.4 and its import price index is 136.9. Calculate the terms of trade index and interpret your answer. (3)

> TOT = export price index ÷ import price index × 100 (1)
> = 128.4 ÷ 136.9 × 100 = **93.79** (1)
> TOT < 100: the terms of trade are unfavourable – SA must export more to buy the same quantity of imports; it is worse off (1)

8.7 A laptop costs US$1 250. Calculate its rand cost when $1 = R17.85 and when $1 = R18.60 (ignore transport costs). What has happened to the rand? (3)

> 1 250 × 17.85 = **R22 312.50** (1)
> 1 250 × 18.60 = **R23 250.00** (1)
> The rand has depreciated – the laptop is R937.50 more expensive (1)

QUESTION 9 / *VRAAG 9* *The Keynesian model: a closed economy* [12]

In the closed economy of Ditsobotla (no government or foreign sector), when total income increases from R800 million to R1 200 million, household consumption increases from R740 million to R1 060 million. Investment is autonomous at R180 million.

9.1 Calculate the marginal propensity to consume (MPC). (2)

> c = ΔC ÷ ΔY = (1 060 − 740) ÷ (1 200 − 800) (1)
> = 320 ÷ 400 = **0.8** (1)

9.2 Calculate the marginal propensity to save (MPS). (1)

> s = 1 − c = 1 − 0.8 = **0.2** (1)

9.3 Derive the consumption function. (2)

> $C = \bar{C} + cY \rightarrow 740 = \bar{C} + 0.8(800)$ (1)
> $\bar{C} = 740 - 640 = 100$, so **$C = 100 + 0.8Y$** (1)

9.4 Derive the savings function and calculate savings when income is R1 200 million. (2)

> $S = -\bar{C} + sY =$ **$-100 + 0.2Y$** (1)
> $S = -100 + 0.2(1\,200) =$ **R140 million** (check: 1 200 − 1 060 = 140) (1)

9.5 Calculate the multiplier and equilibrium income. (3)

> $\alpha = 1 \div (1 - c) = 1 \div 0.2 =$ **5** (1)
> $Y = \alpha(\bar{C} + \bar{I}) = 5 \times (100 + 180)$ (1)
> = **R1 400 million** (check: $S = -100 + 0.2 \times 1\,400 = 180 = I$) (1)

9.6 Investment increases by R40 million. Calculate the change in equilibrium income and the new equilibrium income. (2)

> $\Delta Y = \alpha \times \Delta I = 5 \times 40 =$ **R200 million** (1)
> $Y1 = 1\,400 + 200 =$ **R1 600 million** (1)

QUESTION 10 / *VRAAG 10* *The Keynesian model: an open economy with government* [18]

The following information describes the economy of Thabong (R million):

$C = 120 + 0.85Y_d$ $t = 0.2$ $I = 260$ $G = 310$ $X = 240$ $Z = 50 + 0.08Y$

(Y_d = disposable income = $(1 - t)Y$.)

10.1 Calculate the value of the multiplier. (3)

> $\alpha = 1 \div [1 - c(1 - t) + m]$ (1)
> $= 1 \div [1 - 0.85(0.8) + 0.08] = 1 \div [1 - 0.68 + 0.08]$ (1)
> $= 1 \div 0.40 =$ **2.50** (1)

10.2 Use the multiplier to calculate equilibrium income. (2)

> $\bar{A} = 120 + 260 + 310 + 240 - 50 = 880$ (1)
> $Y = \alpha \times \bar{A} = 2.5 \times 880 =$ **R2 200 million** (1)

10.3 Calculate equilibrium income using the total (aggregate) expenditure approach. (3)

> $Y = C + I + G + X - Z$ (½)
> $Y = 120 + 0.85(1 - 0.2)Y + 260 + 310 + 240 - 50 - 0.08Y$ (1)
> $Y = 880 + 0.68Y - 0.08Y \rightarrow Y - 0.6Y = 880 \rightarrow 0.4Y = 880$ (1)
> $Y =$ **R2 200 million** (½)

10.4 Calculate the government's tax revenue and its budget balance. Is it a surplus or a deficit? (2)

> T = tY = 0.2 × 2 200 = **R440 million** (1)
> T − G = 440 − 310 = **R130 million surplus** (1)

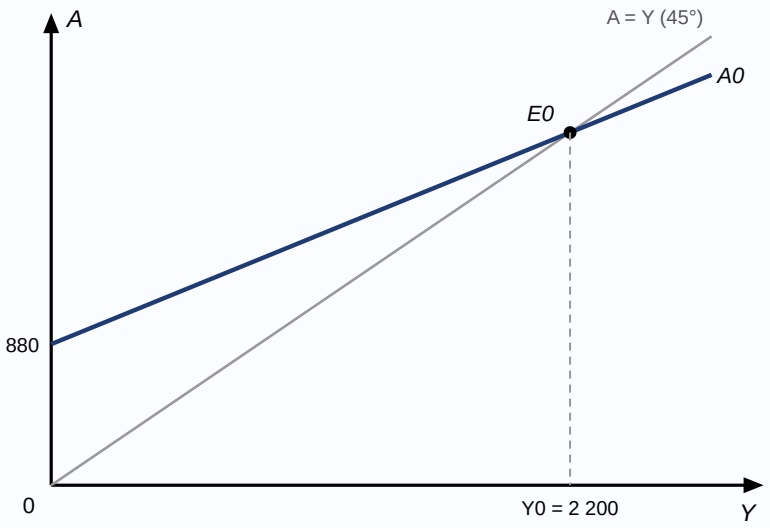
10.5 Calculate total household consumption at equilibrium. (2)

> C = 120 + 0.85(0.8)(2 200) (1)
> = 120 + 1 496 = **R1 616 million** (1)

10.6 The government introduces a basic income grant that increases government spending by R60 million, financed by raising the tax rate to 0.25. Calculate the new equilibrium income using the total expenditure approach. (4)

> Y = 120 + 0.85(1 − 0.25)Y + 260 + 370 + 240 − 50 − 0.08Y (1)
> Y = 940 + 0.6375Y − 0.08Y (1)
> Y − 0.5575Y = 940 → 0.4425Y = 940 (1)
> Y = **R2 124.29 million** (income falls by R75.71 m because the higher tax rate reduces the multiplier to 2.26) (1)

10.7 Illustrate your answer in Question 10.2 graphically. Indicate the intercept, the 45° line and the equilibrium point. (2)



A = 880 + 0.6Y cuts the vertical axis at 880 (½); 45° line (½); equilibrium E0 at Y = 2 200 where A = Y (1).

END OF MEMORANDUM